

I network

- **AI-Powered Network Management Solution**



- **INETWORK leverages advanced artificial intelligence to continuously monitor network traffic patterns, predict congestion points before they impact performance, and automatically implement dynamic bandwidth reallocation strategies**



OUR APPROACH

- **A two-stage LSTM-based solution combining congestion classification with intelligent traffic forecasting and dynamic bandwidth reallocation.**



PRIMARY OBJECTIVE

- **Develop real-time network traffic monitoring and analysis system**



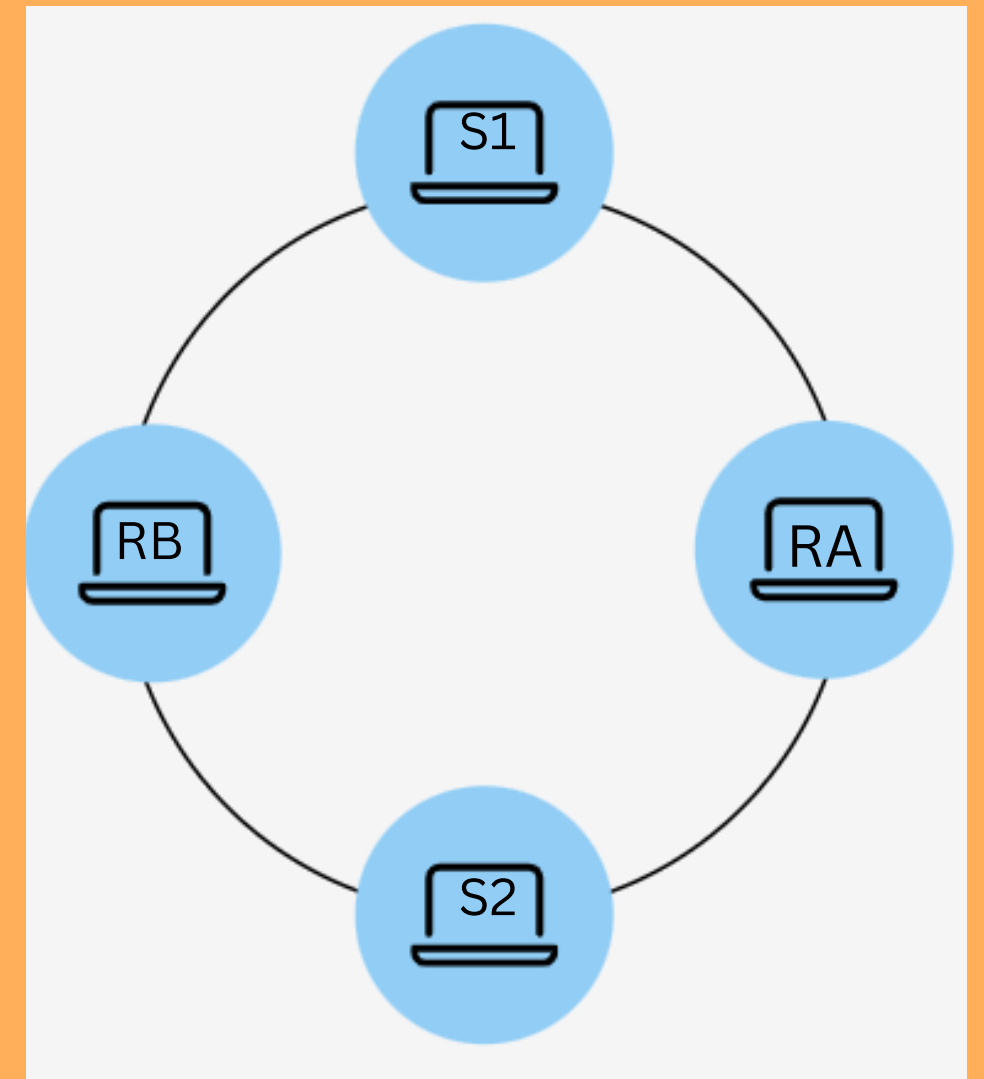
- **Implement predictive congestion detection using machine learning**
- **Create dynamic bandwidth reallocation mechanism**



**"The best way to predict the
future is to create it." - Peter
Drucker**

Network Topology

- **Infrastructure: 2 Routers, 2 Servers architecture**
- **Traffic Simulation: Multi-path routing with variable load distribution**
- **Monitoring Points: Strategic placement at critical network junctions**
- **Data Collection: Real-time traffic metrics, latency measurements, bandwidth utilization**



Models we used

we used the model LSTM

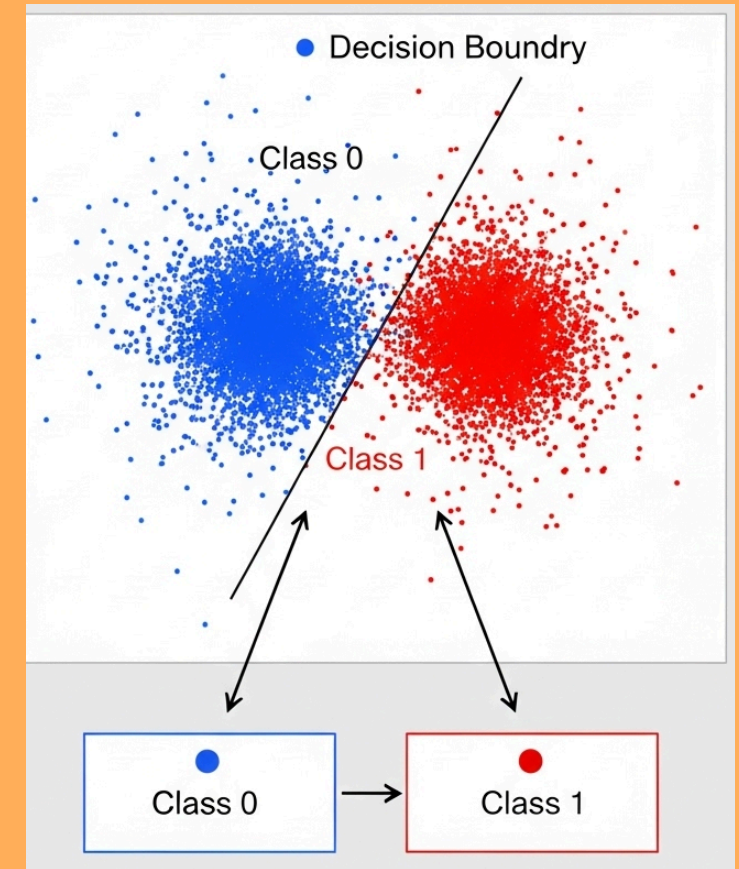
WHY LSTM?:

- **GOOD AT HANDLING NON LINEAR DATA**
- **CAPTURES LONG TERM DEPENDENCIES**

ALGORITHMS USED

STAGE 1

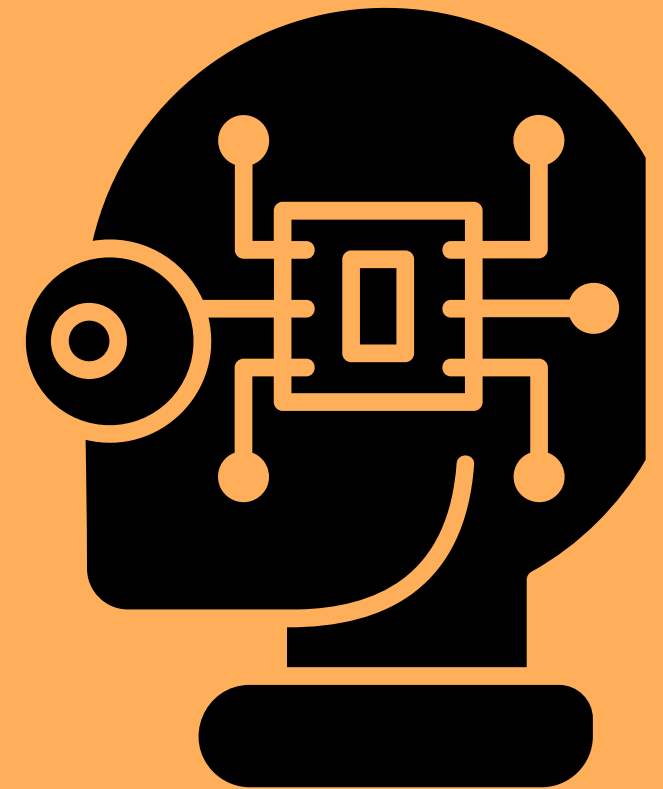
- **LSTM Neural Network for binary classification (Congestion: 0/1)**
- **Historical data pattern recognition**



ALGORITHMS USED

STAGE 2

- **Traffic Forecasting & Response**
- **If Congestion Detected (1): Real-time Reallocation Algorithm (REAL)**
- **If No Congestion (0): LSTM-based time series forecasting for predictive management**



REAL algorthim

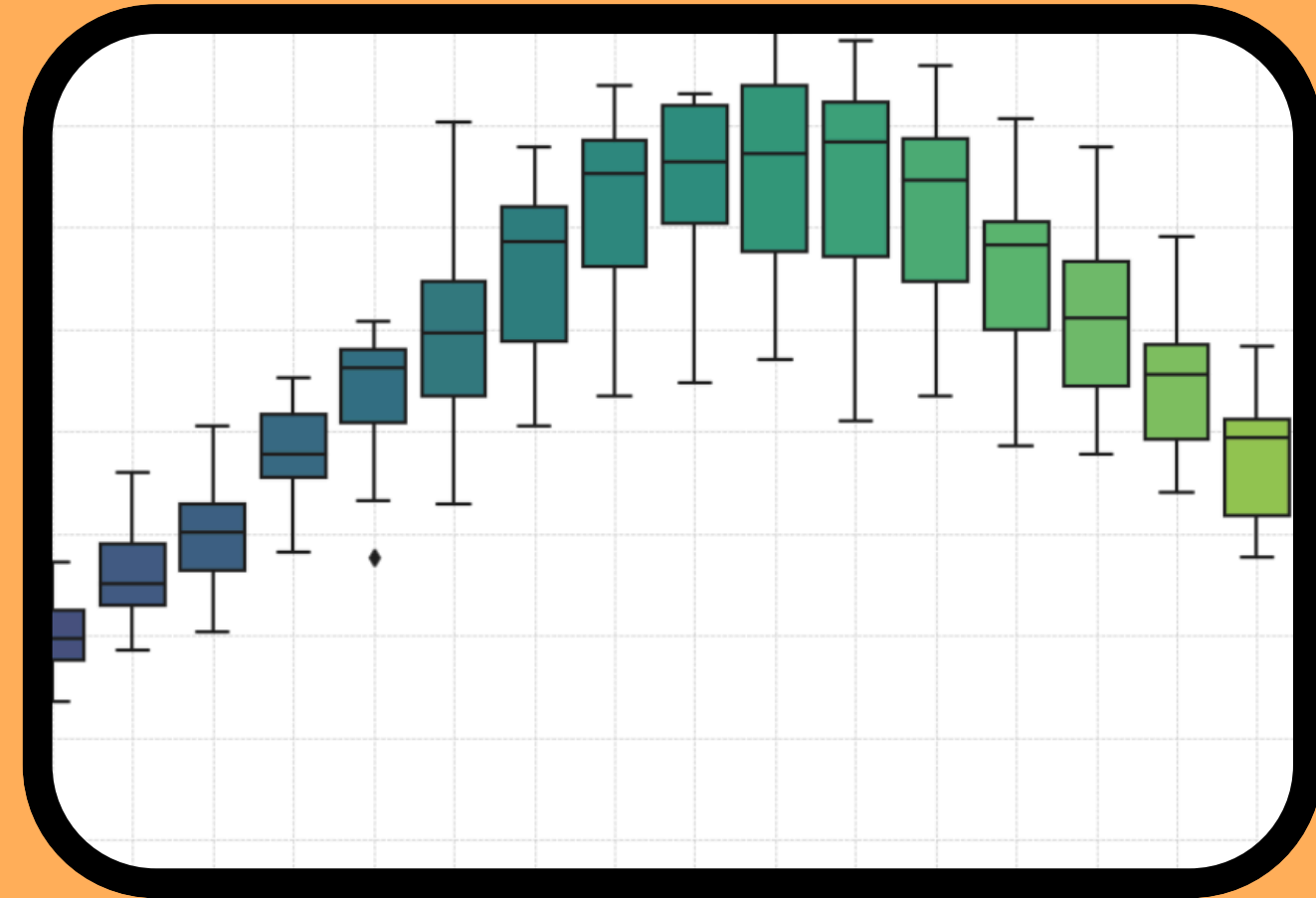
**Dynamic bandwidth
redistribution**

Priority-based traffic routing

Adaptive load balancing

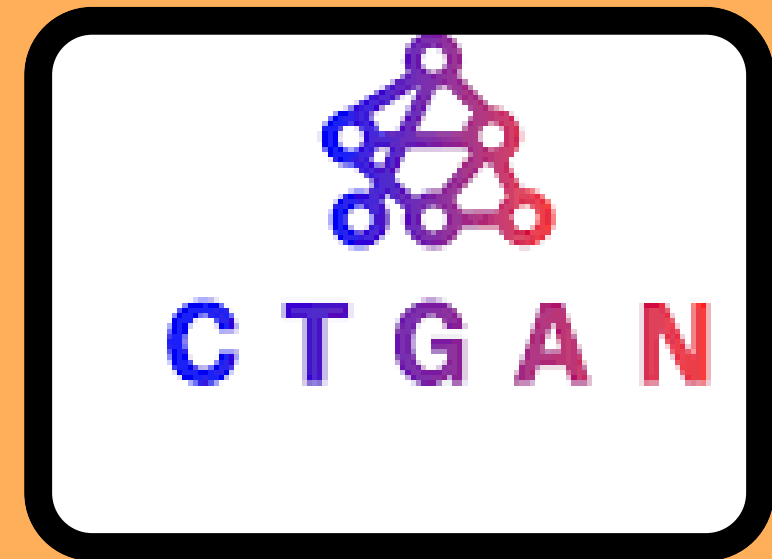
EDA Analysis

- **Data Characteristics:**
- **Non-linear traffic patterns requiring advanced modeling**
- **Tabular format with temporal dependencies**
- **High variability during peak usage periods**



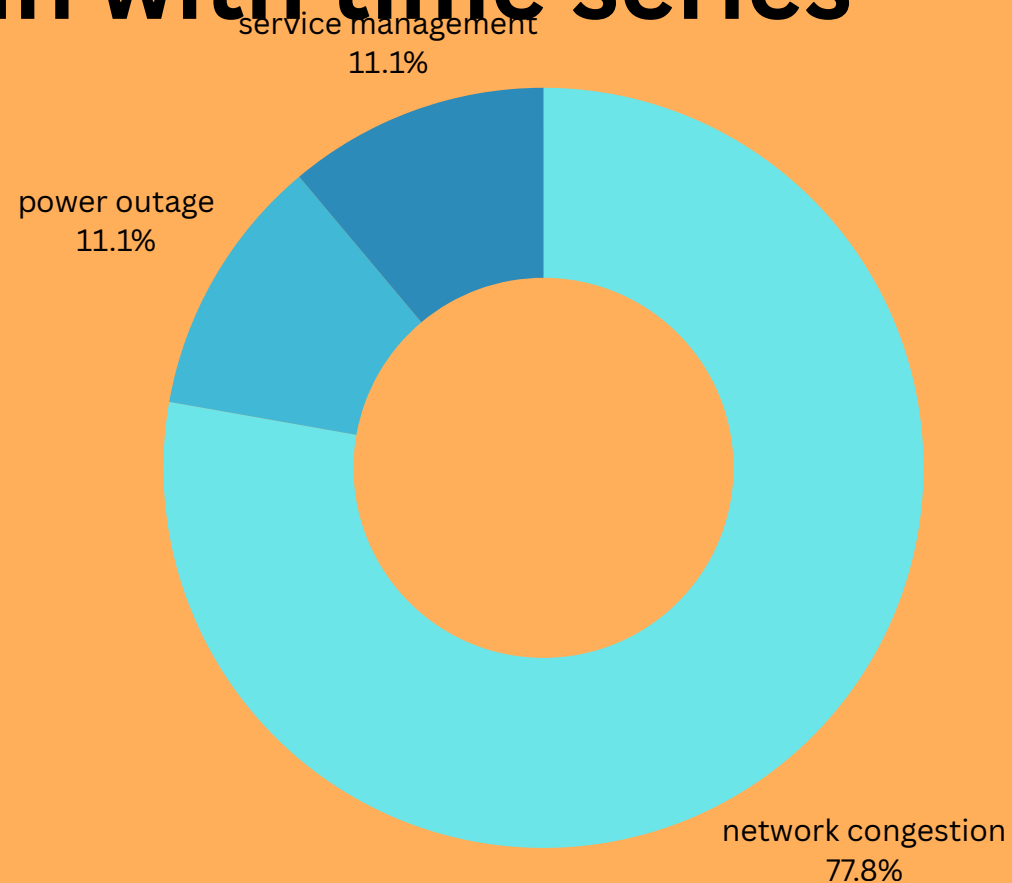
Synthetic Data Generation

- **CTGAN Implementation:**
- **Why CTGAN:**
- **Optimal for non-linear, tabular network data**
- **Preserves complex relationships between network variables**
- **Generates realistic traffic scenarios for model training**
- **Benefits:**
- **Enhanced model robustness through diverse training scenarios**
- **Improved prediction accuracy for edge cases**
- **Reduced dependency on limited historical data**

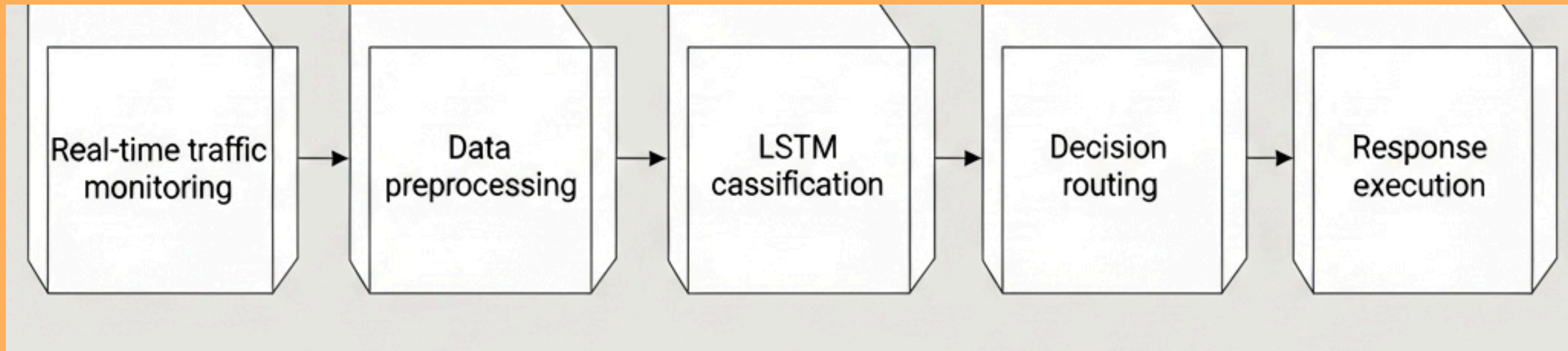


Anomaly Detection Framework

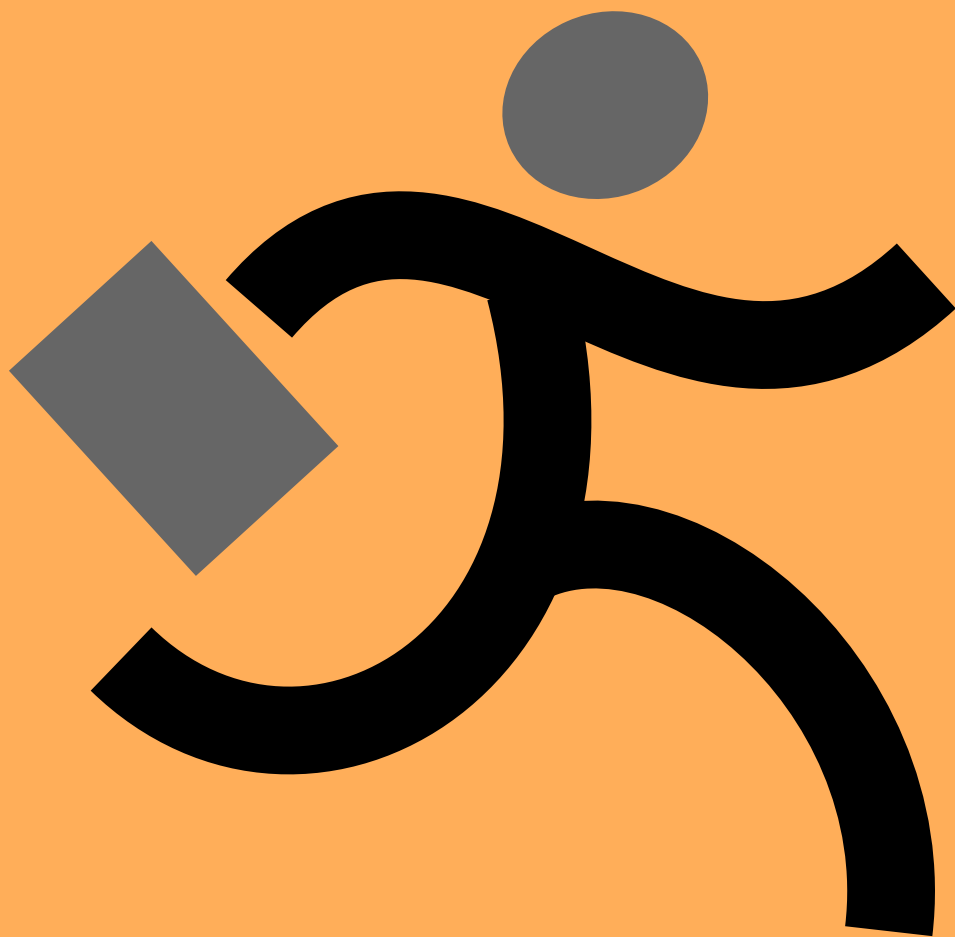
- we labbeled all anoloies as 1,2 ,3
- we then extended our data to fit for 15 days
- we then did classification given this data
- Integration with REAL algorithm for immediate corrective action
- we modelled again with time series



implementation Of our solution



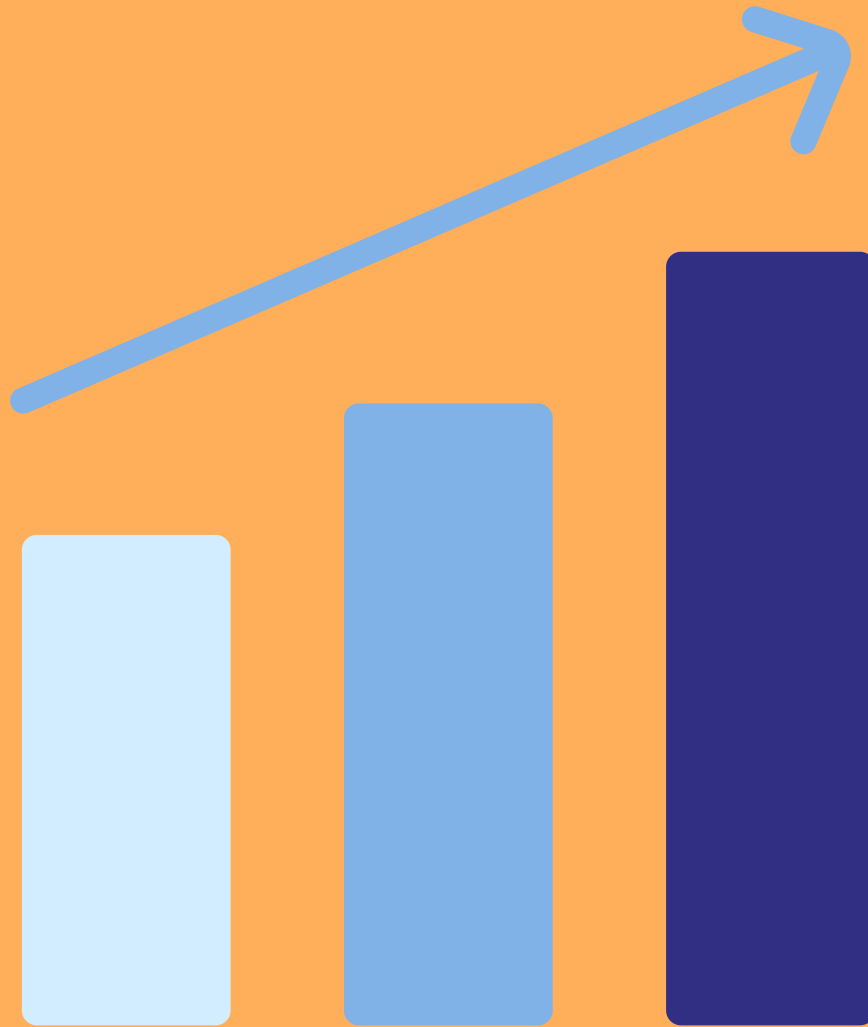
- **Technology Stack:**
- **Python-based machine learning pipeline**
- **LSTM networks using TensorFlow/PyTorch**
- **Real-time data processing capabilities**
- **Automated response system integration**
- **Deployment Features:**
- **Continuous learning from new traffic patterns**
- **Scalable architecture for enterprise environments**
- **Low-latency response for critical network event**



Potential Draw backs

- **High Data Requirements:**
- **LSTM models require substantial historical data for optimal performance**
- **Model accuracy depends on comprehensive traffic pattern coverage**
- **Topology Dependency:**
- **Current solution optimized for specific network configurations**
- **Scaling requires recalibration for different enterprise topologies**

scaling of solution



- **Technology Enhancement:**
- **Google BigQuery:** For large-scale data warehousing and analytics
- **Vertex AI:** For managed machine learning operations and model deployment
- **Cloud-native Architecture:** For automatic scaling and resource optimization

BUISNESS IMPACT AND ROI



- **Cost Reduction:**
- **30-40% reduction in network downtime incidents**
- **Performance Improvements:**
- **Enhanced employee productivity through reliable connectivity**
- **Improved customer experience for client-facing applications**

THANK YOU!

